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1 Joins

Joins are an extremely powerful tool for data science, allowing us to combine two datasets by matching columns.

Some common scenarios:

- Data from two different sources for the same units
  - Pre- and post-treatment
  - Multi-round experiments / survey panels
  - Advertising and vote share
- Data on different levels
  - Students in classrooms in schools
  - Census block groups in counties in states

1.1 Types of joins

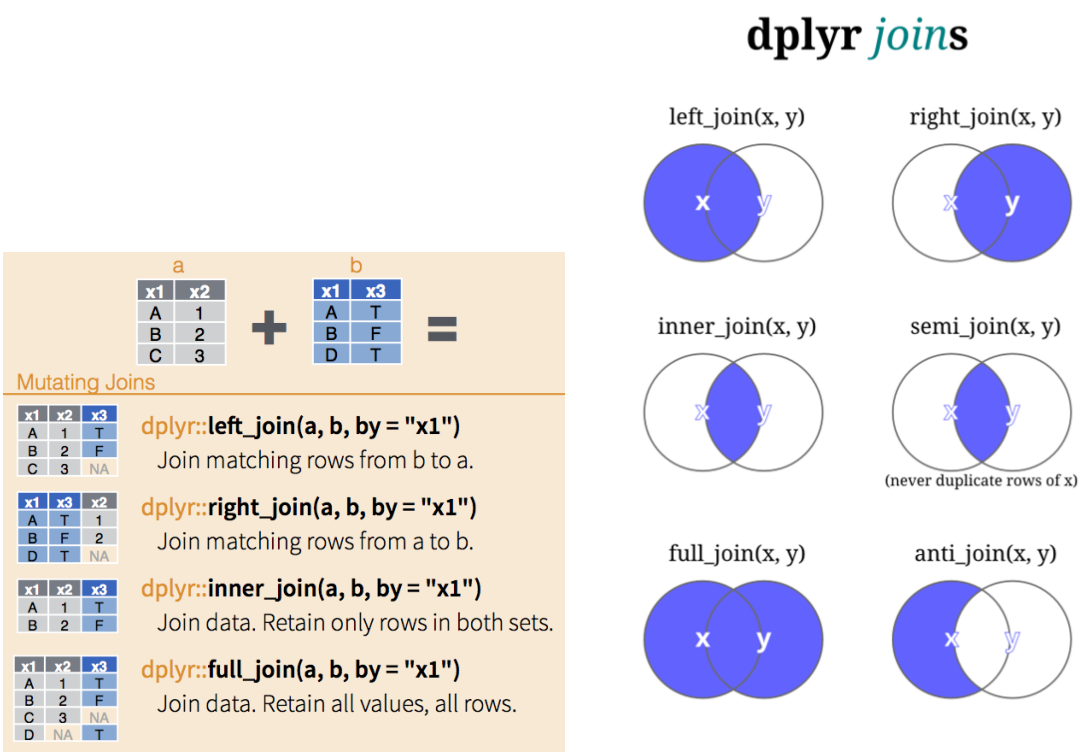

Figure 1: Join operations combine two dataframes based on one or more matching columns. Different join operations keep different rows from the combination. Sources: <https://data-lessons.github.io/gapminder-R/>; <https://notchained.hatenablog.com/entry/2015/01/03/160700>

- Suppose we have two dataframes, `left` and `right`

```r
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.0 --
```

```
## v ggplot2 3.3.2     v purrr   0.3.4
## v tibble  3.0.3     v dplyr   1.0.2
## v tidyr   1.1.2     v stringr 1.4.0
## v readr   1.3.1     v forcats 0.5.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```r
left = tribble(
    ~animal, ~zoo,
    'aardvark', 'Sacramento',
    'aardvark', 'San Diego',
    'chimpanzee', 'San Diego',
    'zebra', 'San Diego'
)
right = tribble(
    ~animal, ~food,
    'aardvark', 'insects',
    'chimpanzee', 'insects',
    'chimpanzee', 'fruit',
    'velociraptor', 'animals'
)
```

- *Left join*: keep all of the rows and columns from `left`; add columns from `right`

```r
left_join(left, right, by = 'animal')
```

```
## # A tibble: 5 x 3
##   animal     zoo        food
##   <chr>      <chr>      <chr>
## 1 aardvark   Sacramento insects
## 2 aardvark   San Diego  insects
## 3 chimpanzee San Diego  insects
## 4 chimpanzee San Diego  fruit
## 5 zebra      San Diego  <NA>
```

- *Right join*: keep all of the rows and columns from `right`; add columns from `left`

```r
right_join(left, right, by = 'animal')
```

```
## # A tibble: 5 x 3
##   animal       zoo        food
##   <chr>        <chr>      <chr>
## 1 aardvark     Sacramento insects
## 2 aardvark     San Diego  insects
## 3 chimpanzee   San Diego  insects
## 4 chimpanzee   San Diego  fruit
## 5 velociraptor <NA>       animals
```

- *Full join*: keep all rows and columns from both `left` and `right`

```r
full_join(left, right, by = 'animal')
```

```
## # A tibble: 6 x 3
```

```
##   animal       zoo        food
##   <chr>        <chr>      <chr>
## 1 aardvark     Sacramento insects
## 2 aardvark     San Diego  insects
## 3 chimpanzee   San Diego  insects
## 4 chimpanzee   San Diego  fruit
## 5 zebra        San Diego  <NA>
## 6 velociraptor <NA>       animals
```

- *Inner join*: keep all columns, but only rows that match

```r
inner_join(left, right, by = 'animal')
```

```
## # A tibble: 4 x 3
##   animal     zoo        food
##   <chr>      <chr>      <chr>
## 1 aardvark   Sacramento insects
## 2 aardvark   San Diego  insects
## 3 chimpanzee San Diego  insects
## 4 chimpanzee San Diego  fruit
```

- *Anti-join*: keep all columns from `left`, but only the rows that *don't* match

```r
anti_join(left, right, by = 'animal')
```

```
## # A tibble: 1 x 2
##   animal zoo
##   <chr>  <chr>
## 1 zebra  San Diego
```

1.2 Hazards

Joins can create a number of hazards.

- *Missing values* from left, right, and full joins
  - In a left join, rows from `left` that don't match will have missing values in the columns from `right`
- *Row counts* can be difficult to predict in advance
  - Rows will be copied if they match multiple times
- *Column names* can be changed if they're repeated across dataframes but not used for matching

To manage these hazards

- Use `anti_join()` to figure out where missing values will be
- Use `select()` with `distinct()` to check for unexpected duplications
- Use `names()` to check column names

1.2.1 Data

To examine these hazards, we'll use the `nycflights13` dataset, which includes all flights that departed New York City in 2013.

The dataset comprises 5 different tables, which of which can be linked to the others using joins. We'll focus on combining `flights` with `airports` and `weather`.

```r
library(tidyverse)
library(nycflights13)

flights
```

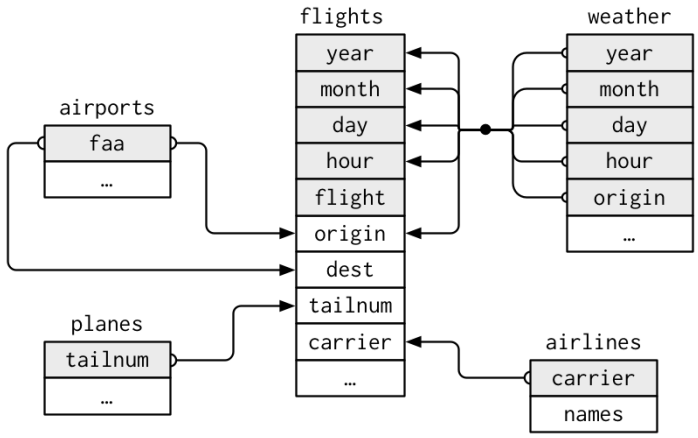

Figure 2: Linking fields in the tables of the `nycflights13` dataset.

```
## # A tibble: 336,776 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      544            545        -1     1004           1022
##  5  2013     1     1      554            600        -6      812            837
##  6  2013     1     1      554            558        -4      740            728
##  7  2013     1     1      555            600        -5      913            854
##  8  2013     1     1      557            600        -3      709            723
##  9  2013     1     1      557            600        -3      838            846
## 10  2013     1     1      558            600        -2      753            745
## # ... with 336,766 more rows, and 11 more variables: arr_delay <dbl>,
## #   carrier <chr>, flight <int>, tailnum <chr>, origin <chr>, dest <chr>,
## #   air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dttm>
```

```
airports
```

```
## # A tibble: 1,458 x 8
##    faa   name                       lat    lon   alt    tz dst   tzone
##    <chr> <chr>                    <dbl>  <dbl> <dbl> <dbl> <chr> <chr>
##  1 04G   Lansdowne Airport         41.1  -80.6  1044    -5 A     America/New_Yo~
##  2 06A   Moton Field Municipal A~  32.5  -85.7   264    -6 A     America/Chicago
##  3 06C   Schaumburg Regional       42.0  -88.1   801    -6 A     America/Chicago
##  4 06N   Randall Airport           41.4  -74.4   523    -5 A     America/New_Yo~
##  5 09J   Jekyll Island Airport     31.1  -81.4    11    -5 A     America/New_Yo~
##  6 0A9   Elizabethton Municipal ~  36.4  -82.2  1593    -5 A     America/New_Yo~
##  7 0G6   Williams County Airport   41.5  -84.5   730    -5 A     America/New_Yo~
##  8 0G7   Finger Lakes Regional A~  42.9  -76.8   492    -5 A     America/New_Yo~
##  9 0P2   Shoestring Aviation Air~  39.8  -76.6  1000    -5 U     America/New_Yo~
## 10 0S9   Jefferson County Intl     48.1 -123.    108    -8 A     America/Los_An~
## # ... with 1,448 more rows
```

```
weather
```

```
## # A tibble: 26,115 x 15
##    origin  year month   day  hour  temp  dewp humid wind_dir wind_speed
```

```
##    <chr>  <int> <int> <int> <int> <dbl> <dbl> <dbl>    <dbl>      <dbl>
##  1 EWR     2013     1     1     1  39.0  26.1  59.4      270      10.4
##  2 EWR     2013     1     1     2  39.0  27.0  61.6      250       8.06
##  3 EWR     2013     1     1     3  39.0  28.0  64.4      240      11.5
##  4 EWR     2013     1     1     4  39.9  28.0  62.2      250      12.7
##  5 EWR     2013     1     1     5  39.0  28.0  64.4      260      12.7
##  6 EWR     2013     1     1     6  37.9  28.0  67.2      240      11.5
##  7 EWR     2013     1     1     7  39.0  28.0  64.4      240      15.0
##  8 EWR     2013     1     1     8  39.9  28.0  62.2      250      10.4
##  9 EWR     2013     1     1     9  39.9  28.0  62.2      260      15.0
## 10 EWR     2013     1     1    10  41    28.0  59.6      260      13.8
## # ... with 26,105 more rows, and 5 more variables: wind_gust <dbl>,
## #   precip <dbl>, pressure <dbl>, visib <dbl>, time_hour <dttm>
```

Some quick data quality observations:

```r
## `flights` has 8.3k missing values for departure and arrival times
tail(flights)
```

```
## # A tibble: 6 x 19
##    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
## 1  2013     9    30       NA           1842        NA       NA           2019
## 2  2013     9    30       NA           1455        NA       NA           1634
## 3  2013     9    30       NA           2200        NA       NA           2312
## 4  2013     9    30       NA           1210        NA       NA           1330
## 5  2013     9    30       NA           1159        NA       NA           1344
## 6  2013     9    30       NA            840        NA       NA           1020
## # ... with 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```r
flights %>%
    filter(is.na(dep_time)) %>%
    nrow()
```

```
## [1] 8255
```

```r
## But origin is complete
flights %>%
    filter(is.na(origin)) %>%
    nrow()
```

```
## [1] 0
```

```r
## And has 3 different values, for the 3 airports in New York City
count(flights, origin)
```

```
## # A tibble: 3 x 2
##   origin      n
##   <chr>   <int>
## 1 EWR    120835
## 2 JFK    111279
## 3 LGA    104662
```

1.2.2 A good join

Expectation: For every flight, we can match the origin code to `airports`.

```r
goodjoin = left_join(flights, airports, by = c('origin' = 'faa'))

nrow(flights)
```

```
## [1] 336776
```

```r
nrow(goodjoin)
```

```
## [1] 336776
```

```r
## If we couldn't match an origin code, `tzone` would be NA
goodjoin %>%
    filter(is.na(tzone))
```

```
## # A tibble: 0 x 26
## # ... with 26 variables: year <int>, month <int>, day <int>, dep_time <int>,
## #   sched_dep_time <int>, dep_delay <dbl>, arr_time <int>,
## #   sched_arr_time <int>, arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>, name <chr>, lat <dbl>,
## #   lon <dbl>, alt <dbl>, tz <dbl>, dst <chr>, tzone <chr>
```

1.2.3 Bad join #1

Expectation: Every airport has at least 1 originating flight.

```r
problems = right_join(flights, airports, by = c('origin' = 'faa'))

## Row counts
nrow(flights)
```

```
## [1] 336776
```

```r
nrow(problems)
```

```
## [1] 338231
```

```r
## Missing values
tail(problems)
```

```
## # A tibble: 6 x 26
##    year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
## 1    NA    NA    NA       NA             NA        NA       NA             NA
## 2    NA    NA    NA       NA             NA        NA       NA             NA
## 3    NA    NA    NA       NA             NA        NA       NA             NA
## 4    NA    NA    NA       NA             NA        NA       NA             NA
## 5    NA    NA    NA       NA             NA        NA       NA             NA
## 6    NA    NA    NA       NA             NA        NA       NA             NA
## # ... with 18 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>, name <chr>, lat <dbl>,
## #   lon <dbl>, alt <dbl>, tz <dbl>, dst <chr>, tzone <chr>
```

```r
problems %>%
    filter(is.na(carrier)) %>%
    nrow()
```

```
## [1] 1455
```

```r
problems %>%
    filter(is.na(carrier)) %>%
    select(year, month, day, carrier, origin, name, tzone)
```

```
## # A tibble: 1,455 x 7
##     year month   day carrier origin name                        tzone
##    <int> <int> <int> <chr>   <chr>  <chr>                       <chr>
##  1    NA    NA    NA <NA>    04G    Lansdowne Airport           America/New_York
##  2    NA    NA    NA <NA>    06A    Moton Field Municipal Airp~ America/Chicago
##  3    NA    NA    NA <NA>    06C    Schaumburg Regional         America/Chicago
##  4    NA    NA    NA <NA>    06N    Randall Airport             America/New_York
##  5    NA    NA    NA <NA>    09J    Jekyll Island Airport       America/New_York
##  6    NA    NA    NA <NA>    0A9    Elizabethton Municipal Air~ America/New_York
##  7    NA    NA    NA <NA>    0G6    Williams County Airport     America/New_York
##  8    NA    NA    NA <NA>    0G7    Finger Lakes Regional Airp~ America/New_York
##  9    NA    NA    NA <NA>    0P2    Shoestring Aviation Airfie~ America/New_York
## 10    NA    NA    NA <NA>    0S9    Jefferson County Intl       America/Los_Ang~
## # ... with 1,445 more rows
```

Manage this using `anti_join()` to determine which airports aren't in `origin`.

```r
anti_join(airports, flights, by = c('faa' = 'origin'))
```

```
## # A tibble: 1,455 x 8
##    faa   name                       lat    lon   alt    tz dst   tzone
##    <chr> <chr>                    <dbl>  <dbl> <dbl> <dbl> <chr> <chr>
##  1 04G   Lansdowne Airport         41.1  -80.6  1044    -5 A     America/New_Yo~
##  2 06A   Moton Field Municipal A~  32.5  -85.7   264    -6 A     America/Chicago
##  3 06C   Schaumburg Regional       42.0  -88.1   801    -6 A     America/Chicago
##  4 06N   Randall Airport           41.4  -74.4   523    -5 A     America/New_Yo~
##  5 09J   Jekyll Island Airport     31.1  -81.4    11    -5 A     America/New_Yo~
##  6 0A9   Elizabethton Municipal ~  36.4  -82.2  1593    -5 A     America/New_Yo~
##  7 0G6   Williams County Airport   41.5  -84.5   730    -5 A     America/New_Yo~
##  8 0G7   Finger Lakes Regional A~  42.9  -76.8   492    -5 A     America/New_Yo~
##  9 0P2   Shoestring Aviation Air~  39.8  -76.6  1000    -5 U     America/New_Yo~
## 10 0S9   Jefferson County Intl     48.1 -123.    108    -8 A     America/Los_An~
## # ... with 1,445 more rows
```

1.2.4 Bad join #2

Calculate the average daily temperature at each airport.

```r
avg_temp = weather %>%
    group_by(origin, year, month, day) %>%
    summarize(temp = mean(temp)) %>%
    ungroup()
```

```
## `summarise()` regrouping output by 'origin', 'year', 'month' (override with `.groups` argument)
```

Expectation: Each flight has 1 average daily temperature at its origin.

```r
mult_problems = inner_join(flights, avg_temp,
                           by = c('year', 'month', 'day'))
nrow(flights)
```

```
## [1] 336776
```

```r
nrow(mult_problems)
```

```
## [1] 1008000
```

```r
## `mult_problems` has almost 3x more rows than `flights`
nrow(mult_problems) / nrow(flights)
```

```
## [1] 2.993087
```

One way to figure out what happened is to check the names of the columns. The `dplyr` join functions add `.x` and `.y` when they encounter repeated column names.

```r
mult_problems %>%
    select(matches('.x'))
```

```
## # A tibble: 1,008,000 x 1
##    origin.x
##    <chr>
##  1 EWR
##  2 EWR
##  3 EWR
##  4 LGA
##  5 LGA
##  6 LGA
##  7 JFK
##  8 JFK
##  9 JFK
## 10 JFK
## # ... with 1,007,990 more rows
```

```r
count(mult_problems, origin.x, origin.y)
```

```
## # A tibble: 9 x 3
##   origin.x origin.y      n
##   <chr>    <chr>     <int>
## 1 EWR      EWR      120565
## 2 EWR      JFK      120565
## 3 EWR      LGA      120565
## 4 JFK      EWR      110996
## 5 JFK      JFK      110996
## 6 JFK      LGA      110996
## 7 LGA      EWR      104439
## 8 LGA      JFK      104439
## 9 LGA      LGA      104439
```